

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1003	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Code of Conduct

I _____PRIYADHARSHINI K S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ PRIYADHARSHINI K S _____

Vehicle Service Scheduling and Predictive Maintenance Management System

1. Study and Analyse the Scenario

- Carefully understand the given problem statement.

The **Vehicle Service Scheduling and Predictive Maintenance Management System** is proposed to overcome the difficulties involved in managing vehicle servicing manually. Vehicle owners often face problems such as missing service schedules, delayed maintenance, and lack of proper service-history information. Service centres also need to manage appointments, technicians, vehicle inspections, repairs, and spare parts efficiently. The proposed system provides a centralized platform for scheduling vehicle services and maintaining digital service records. It also analyses vehicle usage and previous maintenance information to predict upcoming maintenance requirements. Therefore, the system aims to provide timely servicing, improve maintenance efficiency, reduce unexpected vehicle failures, and improve customer satisfaction.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain:

The proposed system belongs to the automotive service and vehicle maintenance domain. It is developed to automate and manage vehicle servicing activities through a centralized software platform. The system allows vehicle owners to schedule services, maintain vehicle records, receive maintenance reminders, and monitor service status. Service centres can manage appointments, technicians, service records, spare parts, and billing. The system also uses historical vehicle information and service data to predict future maintenance requirements. Thus, the system combines vehicle service scheduling and predictive maintenance management.

Stakeholders:

The major stakeholders of the system are:

1. **Vehicle Owner/Customer** – Books services and views vehicle information.
2. **Service Advisor** – Manages appointments and customer service requests.
3. **Mechanic/Technician** – Inspects vehicles and performs maintenance.
4. **Service Centre Manager** – Monitors overall service operations.
5. **System Administrator** – Manages users, roles, and system settings.
6. **Spare Parts Manager** – Manages spare-parts inventory.
7. **Maintenance Management Team** – Monitors maintenance predictions and recommendations.
8. **Payment Service Provider** – Handles online service payments.
9. **Notification Service** – Sends reminders and maintenance alerts.
10. **Business Owner/Management** – Uses reports for operational decisions.

Existing Challenges

Traditional vehicle servicing often depends on manual records, telephone calls, spreadsheets, and paper documents. Customers may forget their scheduled maintenance and may not know when important components require inspection or replacement. Service centres can face appointment conflicts, inefficient technician allocation, missing service records, and spare-parts availability problems. It is also difficult to identify possible future vehicle problems using only manual methods. These issues can lead to delayed servicing, increased maintenance costs, and unexpected vehicle breakdowns. Therefore, an automated system is required to manage service scheduling and predictive maintenance efficiently.

System Objectives:

The major objectives of the proposed system are:

1. To provide an online vehicle service booking facility.
2. To maintain digital vehicle and service records.
3. To reduce manual service scheduling activities.
4. To send automatic service reminders to customers.
5. To predict future vehicle maintenance requirements.
6. To identify possible maintenance problems at an early stage.
7. To efficiently assign service tasks to technicians.

8. To manage spare-parts availability and usage.
9. To provide service status and maintenance reports.
10. To improve customer satisfaction and reduce unexpected breakdowns.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

User Registration and Login: The system shall allow customers, service advisors, technicians, managers, and administrators to register or log in securely.

Vehicle Management: The system shall allow customers to add, update, and manage vehicle details such as registration number, model, year, and mileage.

Service Appointment Booking: The system shall allow customers to select a required service and book an available date and time slot.

Appointment Management: The system shall allow service advisors to confirm, reschedule, or cancel customer service appointments.

Technician Assignment: The system shall allow service advisors or managers to assign service jobs to available technicians based on service requirements.

Vehicle Inspection and Service: The system shall allow technicians to record vehicle inspection results, repair details, maintenance activities, and service completion status.

Service History Management: The system shall maintain a digital history of all services, repairs, replaced parts, mileage, and maintenance activities for each vehicle.

Predictive Maintenance: The system shall analyse vehicle usage, mileage, and previous service records to identify and predict upcoming maintenance requirements.

Notifications and Reminders: The system shall send service reminders, appointment confirmations, maintenance alerts, and service-completion notifications to customers.

Billing and Reports: The system shall generate service invoices and provide reports related to appointments, vehicle maintenance, services, spare parts, and payments.

- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
 1. **Performance:** The system should respond quickly to operations such as user login, vehicle search, appointment booking, and service-history retrieval without unnecessary delays.
 2. **Security:** The system must protect customer, vehicle, service, and payment information through secure login, authentication, authorization, and appropriate access controls.
 3. **Reliability:** The system should perform its functions correctly and consistently without frequent failures, ensuring that service and maintenance records are not lost or corrupted.
 4. **Usability:** The system should provide a simple, user-friendly interface so that customers, technicians, service advisors, and administrators can easily use the required functions.
 5. **Scalability:** The system should be capable of supporting an increasing number of customers, vehicles, appointments, service records, and service centres without major performance problems.
 6. **Maintainability:** The system should have a modular and well-structured design so that developers can easily identify errors, fix problems, and add new features.
 7. **Availability:** The system should be available whenever customers and service-centre employees need to access appointment, vehicle, and maintenance information.
 8. **Data Integrity:** The system should ensure that vehicle details, appointment information, service history, spare-parts records, and payment information remain accurate and consistent.
 9. **Backup and Recovery:** The system should regularly back up important data and provide recovery mechanisms to prevent permanent loss of vehicle and service information.

10. **Compatibility:** The system should operate correctly on commonly used web browsers, computers, tablets, and smartphones, allowing users to access the system from different devices.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors:

The primary actors interacting directly with the system are:

1. Vehicle Owner/Customer
2. Service Advisor
3. Mechanic/Technician
4. Service Centre Manager
5. System Administrator
6. Spare Parts Manager

Secondary Actors:

The secondary actors are:

1. Payment Gateway
2. Notification Service
3. Predictive Maintenance Engine
4. Database Management System

- List the major use cases associated with each actor.

Actor	Major Use Cases
Customer	Register, Login, Manage Vehicle, Book Service, Cancel Appointment, Track Service, View History

Service Advisor	Manage Appointments, Create Service Request, Assign Technician, Update Status
Mechanic	View Jobs, Inspect Vehicle, Perform Service, Update Repairs, Request Parts
Manager	Monitor Services, Manage Technicians, View Reports, Monitor Performance
Administrator	Manage Users, Manage Roles, Configure System, Generate Reports
Parts Manager	Manage Parts, Update Stock, Process Parts Requests
Payment Gateway	Process Payment, Verify Payment
Notification Service	Send Reminders, Send Alerts
Predictive Engine	Analyse Data, Predict Maintenance, Generate Recommendation

- Draw a Use Case Diagram representing the interactions between actors and the system.

4. **Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

- **Introduction**

The **Vehicle Service Scheduling and Predictive Maintenance Management System** is designed to automate vehicle servicing and maintenance activities. The system provides a centralized platform connecting customers and vehicle service-centre employees.

Customers can register vehicles, schedule appointments, view service history, and receive maintenance notifications. Service-centre staff can manage appointments, technicians, vehicle inspections, repairs, spare parts, and invoices. The predictive maintenance component analyses available vehicle and historical service information to identify possible future maintenance requirements. The system aims to improve service efficiency, reduce manual work, and minimize unexpected vehicle failures.

- **Problem Description**

Vehicle servicing is often managed using manual records, telephone communication, spreadsheets, or paper documents. These methods can result in appointment conflicts, missing service information, delayed maintenance, and inefficient technician allocation. Customers may forget service schedules and may not recognize early signs of vehicle problems. Service centres also face difficulties in managing spare parts and maintaining accurate service histories. Traditional systems provide limited support for predicting future maintenance requirements. The proposed system solves these problems by integrating service scheduling, digital service records, notifications, and predictive maintenance into a single platform.

- **Scope of the System**

The system covers customer management, vehicle registration, service appointment scheduling, technician assignment, vehicle inspection, repair tracking, service history, spare-parts management, billing, notifications, and predictive maintenance. Customers can book and monitor services through the system. Service-centre employees can manage appointments, technicians, repairs, and inventory. The predictive maintenance module uses vehicle and historical service data to generate maintenance recommendations.

Administrators can manage users and system configurations. The system can later be extended with IoT sensors, real-time vehicle data, mobile applications, and advanced machine-learning techniques.

- **Functional Requirements**

The functional requirements can be grouped into modules:

User Management

- User registration
- User login
- Profile management
- Password management
- Role-based access

Vehicle Management

- Vehicle registration
- Vehicle details update
- Mileage update
- Service history
- Vehicle status

Appointment Management

- Service booking
- Available slot checking
- Appointment confirmation
- Rescheduling
- Cancellation

Service Management

- Vehicle inspection
- Technician assignment

- Repair recording
- Service status update
- Service completion

Predictive Maintenance

- Collection of vehicle data
- Analysis of service history
- Maintenance pattern detection
- Maintenance prediction
- Recommendation generation

Inventory Management

- Spare-part registration
- Stock update
- Parts request
- Parts issue
- Low-stock monitoring

Billing

- Service estimate
- Invoice generation
- Payment processing
- Payment status

○ **Non-Functional Requirements**

The system should provide reliable and secure services to all authorized users. It should respond quickly to operations such as login, appointment booking, and service-history retrieval. The system should protect customer and vehicle information using authentication and authorization. It should be scalable enough to support multiple service centres and large numbers of vehicles. The user interface should be simple and accessible to customers

and employees. Regular backups should be maintained to prevent data loss. The system should also be maintainable so that future predictive-maintenance and IoT features can be integrated easily.

- **Use Case Descriptions**

Use Case 1: Customer Books Service

Actor: Customer

Precondition: Customer must have a valid account.

Steps:

1. Customer logs into the system.
2. Customer selects a registered vehicle.
3. Customer selects the required service.
4. System displays available dates and time slots.
5. Customer selects a suitable slot.
6. System checks slot availability.
7. System confirms the appointment.
8. Notification is sent to the customer.

Postcondition: The appointment is successfully stored in the system.

Use Case 2: Technician Performs Service

Actor: Mechanic/Technician

Precondition: A valid service appointment must be assigned to the technician.

Steps:

1. Technician logs into the system.
2. Technician views assigned vehicles.

3. Technician selects a service task.
4. Technician performs vehicle inspection.
5. Technician records identified problems.
6. Technician performs required maintenance.
7. Technician records parts and repair details.
8. Technician marks the service as completed.

Postcondition: The service record is updated and stored.

Use Case 3: Predictive Maintenance

Actor: Predictive Maintenance Engine

Steps:

1. System collects vehicle information.
2. System retrieves historical service records.
3. System analyses mileage and service patterns.
4. Predictive engine processes the available data.
5. Possible maintenance requirements are identified.
6. Maintenance risk is calculated.
7. A maintenance recommendation is generated.
8. Customer receives a maintenance notification.

Postcondition: The prediction and recommendation are stored for future reference

- Data Requirements

Data Entity	Main Attributes
Customer	Customer ID, Name, Phone, Email
Vehicle	Vehicle ID, Registration No., Model, Year, Mileage
Appointment	Appointment ID, Date, Time, Service Type, Status
Service	Service ID, Vehicle ID, Technician ID, Details
Technician	Technician ID, Name, Skill, Availability
Spare Part	Part ID, Name, Quantity, Price
Invoice	Invoice ID, Service ID, Amount, Status
Payment	Payment ID, Invoice ID, Amount, Date
Prediction	Prediction ID, Vehicle ID, Risk, Recommendation
Notification	Notification ID, User ID, Message, Status

- External Interface Requirements

The system requires several external interfaces to provide complete functionality. A **payment gateway** can be used for processing online service payments. A **notification service** can send SMS, email, or application notifications to customers. A database system stores customer, vehicle, appointment, service, inventory, and prediction information. In future versions, vehicle diagnostic devices or IoT sensors can provide real-time vehicle information. APIs can be used to communicate securely with these external services. All external communication should use secure protocols to protect data.

- System Features

Customer Management – Manages customer registration and profiles.

Vehicle Management – Stores vehicle details and service history.

Service Scheduling – Allows customers to book and manage appointments.

Technician Management – Assigns service tasks to technicians.

Service Tracking – Displays the current status of vehicle service.

Predictive Maintenance – Predicts possible future maintenance requirements.

Notification System – Sends reminders and maintenance alerts.

Inventory Management – Tracks spare-parts availability and usage.

Billing and Payment – Generates invoices and manages payments.

Reports and Analytics – Provides useful service and maintenance reports.

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

The system is developed based on the following assumptions:

1. Customers have access to the internet.
2. Customers provide accurate vehicle information.
3. Service-centre employees have valid system accounts.
4. Service centres maintain updated appointment availability.
5. Technicians correctly update service records.
6. Vehicle mileage is updated regularly.
7. Historical maintenance records are available.
8. Notification services are available when required.
9. Payment gateway services are available for online payments.
10. Predictive maintenance recommendations are verified by qualified technicians.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Technical Constraints

1. System performance depends on server and database capacity.
2. Prediction accuracy depends on the quality of historical data.
3. IoT integration may require compatible hardware.
4. Internet connectivity is required for online access.

Operational Constraints

5. Employees must update service information correctly.
6. Staff members require training to use the system.
7. Appointment availability depends on service-centre capacity.

Economic Constraints

8. Advanced predictive analytics may increase development costs.
9. Payment and notification services may involve third-party charges.

Legal Constraints

10. Customer information must be protected according to applicable privacy and data-protection requirements.

Time Constraints

11. Core scheduling features may need to be developed before advanced predictive features.
12. IoT and advanced machine-learning features may be implemented in future version

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

The proposed Vehicle Service Scheduling and Predictive Maintenance Management System has been designed to satisfy the requirements of all identified stakeholders.

Customers can register, log in, register their vehicles, schedule service appointments, track service status, and receive maintenance reminders. Service centre managers can efficiently manage appointments, assign mechanics, and monitor daily service activities. Mechanics can access assigned jobs, update service progress, and maintain accurate service records. Administrators can manage users, maintain the database, and generate reports for effective system management. Therefore, the proposed system successfully fulfills both functional and non-functional requirements while ensuring consistency with stakeholder expectations and project objectives.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

The system requirements were carefully reviewed to ensure they are clear, complete, and consistent. All functional and non-functional requirements are written in a simple and unambiguous manner, reducing the possibility of misunderstanding during development. Duplicate or redundant requirements have been removed to avoid unnecessary complexity. The requirements are consistent with the project objectives and stakeholder expectations, ensuring that no conflicts exist between different system functions. A thorough validation process was carried out to identify any missing requirements, and additional features such as maintenance reminders, service history management, and notification services were included. This validation ensures that the proposed system is complete, reliable, and ready for implementation.

- Suggest improvements to enhance the quality and completeness of the requirement specification.

The quality of the requirement specification can be improved by including more detailed functional and non-functional requirements with clear acceptance criteria. Future requirements such as online payment integration, AI-based predictive maintenance, mobile application support, and cloud-based data storage can be added to enhance system

functionality. Security requirements should be strengthened by specifying data encryption, role-based access control, and regular backup mechanisms. Performance requirements such as response time and system availability can also be clearly defined. Periodic stakeholder reviews and requirement validation should be conducted to ensure the specification remains accurate, complete, and aligned with user needs throughout the development process.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case	Correctly identifies all relevant actors and use cases; use case diagram is complete,	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified;	Actors/use cases missing or incorrect;

Modelling (20%)	accurate, and follows UML standards.		diagram partially correct.	diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

